Protocol design elements and USDM

Data or words?

What USDM holds for objectives, endpoints, eligibility criteria and the schedule

Data | Data in a string | Words

Kerstin Forsberg · the published USDM v4 model and its CDISC example files, read as a knowledge graph

The question

For each design element: which parts are data, and which are still words?

AN OBJECTIVE IN USDM · CDISC PILOT STUDY EXAMPLE

Level: secondary objective

Its 3 endpoints, linked

> “To assess the dose-dependent improvement in behavior. Improved scores on the Revised Neuropsychiatric Inventory (NPI-X) will indicate improvement in these areas.”

What is measured, and how it is judged, sits inside the sentence.

Data

Choices from a list, numbers, links. A system can check and reuse them.

Data in a string

An identifier or a duration written as text. Usable only if you know the convention.

Words

Sentences and written rules. A person has to read them.

Each reading here comes from USDM v4 read as a knowledge graph.

Five design elements: the schedule is data, the rest is mostly words

Data | Data in a string | Words

Objective ↗

- Level
- Its endpoints
- The objective, written

Endpoint ↗

- Level
- The endpoint, written
- Its purpose

Eligibility criterion ↗

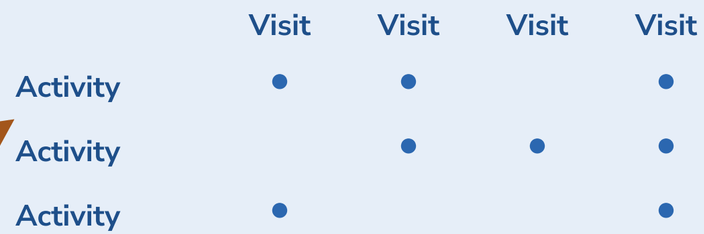
- Inclusion or exclusion
- Order
- Label like “I1”
- The criterion, written

no link in USDM

Protocol document, its sections ↗

no link in USDM

Schedule of activities ↗



- visit type, contact mode
- what is measured
- timing and windows
- Conditions for branching
- Visit start and end rules

Arms × epochs ↗

	Screening	Treatment	Follow-up
Arm 1	element	element	element
Arm 2	element	element	element

Inside each element: what happens, start rule, end rule

In short

USDM gives every design element an identity. Much of the content is still words.

Identity, for the objective on slide 2

USDM, the kind	name `Objective` · C-code `C142450` (Study Objective)
USDM, this one	id `Objective_3` · name `OBJ3` · unique within the study file
usdm-rdf adds	a global identifier for the kind, linked to NCIt C142450: `https://w3id.org/cdisc/usdm/v4/Objective`

Held as data	The schedule of activities, and the arm and epoch grid
Data in a string	Timing values and windows; labels like “I1”
Held as words	Objectives, endpoints, eligibility criteria, rules inside elements
Not held	Endpoint to assessments; design element to protocol section

Beyond USDM

The words are not a dead end

A knowledge graph can represent the entities and relationships from study design to data capture. Structuring what the words say in such a graph turns them into data, and that is where the link from an endpoint to its assessments can live.